Lower bound reasoning with eHL

ANONYMOUS AUTHOR(S)

$$\frac{}{\{\, P \mid f \,\}_{\bowtie}\ \texttt{skip}\ \{\, P \mid f \,\}}\ [\textsc{skip}]$$

$$\frac{}{\{\, P[\mathsf{x}/\mathsf{E}] \mid f[\mathsf{x}/\mathsf{E}] \,\}_{\bowtie}\ \mathsf{x} \leftarrow \mathsf{E}\ \{\, P \mid f \,\}}\ [\textsc{assign}]$$

$$\frac{}{\{\, \forall v \in \mathrm{supp}(\mathsf{S}),\ P[\mathsf{x}/v] \mid \mathbb{E}_{\mathsf{S}}[\lambda v.\, f[\mathsf{x}/v]] \,\}_{\bowtie}\ \mathsf{x} \xleftarrow{\$} \mathsf{S}\ \{\, P \mid f \,\}}\ [\textsc{sample}]$$

$$\frac{\{\, P \mid f \,\}_{\bowtie}\ \mathsf{C}\ \{\, R \mid h \,\} \quad \{\, R \mid h \,\}_{\bowtie}\ \mathsf{D}\ \{\, Q \mid g \,\}}{\{\, P \mid f \,\}_{\bowtie}\ \mathsf{C};\ \mathsf{D}\ \{\, Q \mid g \,\}}\ [\textsc{seq}]$$

$$\frac{\{\, \mathsf{B} \wedge P \mid f \,\}_{\bowtie}\ \mathsf{C}\ \{\, Q \mid g \,\} \quad \{\, \neg\mathsf{B} \wedge P \mid f \,\}_{\bowtie}\ \mathsf{D}\ \{\, Q \mid g \,\}}{\{\, P \mid f \,\}_{\bowtie}\ \texttt{if}\ \mathsf{B}\ \texttt{then}\ \mathsf{C}\ \texttt{else}\ \mathsf{D}\ \{\, Q \mid g \,\}}\ [\textsc{if}]$$

$$\frac{\mathsf{B} \wedge P \vDash f_0 \bowtie 0 \quad \forall k.\, \{\, \mathsf{B} \wedge P \mid f_{k+1} \,\}_{\bowtie}\ \mathsf{C}\ \{\, P \mid f_k \,\} \quad \neg\mathsf{B} \wedge P \vDash f_k \bowtie g}{\{\, P \mid \mathrm{lub} f_k \,\}_{\bowtie}\ \texttt{while}\ \mathsf{B}\ \texttt{do}\ \mathsf{C}\ \{\, \neg\mathsf{B} \wedge P \mid g \,\}}\ [\textsc{while}]$$

Fig. 1. eHL for lower and upper-bounds. Here $\bowtie \in \{\leq, \geq\}$.

$$\frac{\{\, \mathsf{B} \wedge P \mid f \,\}_{\leq}\ \mathsf{C}\ \{\, P \mid f \,\} \quad \mathsf{B} \wedge P \vDash r_0 \geq f \quad \forall i \geq 0.\, \{\, \mathsf{B} \wedge P \mid r_{i+1} \,\}_{\geq}\ \mathsf{C}\ \{\, \top \mid [P] \cdot r_i \,\} \quad \mathsf{B} \wedge P \vDash \lim_{i\to\infty} r_i = 0}{\{\, P \mid f \,\}_{\leq}\ \texttt{while}\ \mathsf{B}\ \texttt{do}\ \mathsf{C}\ \{\, \neg\mathsf{B} \wedge P \mid f \,\}}\ [\textsc{while}]$$

Fig. 2. The While rule for eHL lower bounds. The rule is specific to the lower-bound judgment: Premise 1 is a lower-bound triple on the body, while Premise 3 is an *upper*-bound triple used purely to bound the in-loop mass. The sequence $\{r_i\}_{i\geq 0}$ consists of non-negative expectations $r_i : \mathrm{State} \to \mathbb{R}^+$.

Notation. For a predicate P we use two indicator-style operators, one tailored to each direction of reasoning:

$$[P] \;=\; \begin{cases} 1 & \text{if } P \\ 0 & \text{otherwise} \end{cases} \qquad \text{(used for lower bounds)}$$

$$\langle P \rangle \;=\; \begin{cases} 1 & \text{if } P \\ \infty & \text{otherwise} \end{cases} \qquad \text{(used for upper bounds)}$$

Given a (state-indexed) expectation r, the expressions $[P] \cdot r$ and $\langle P \rangle \cdot r$ denote the corresponding pointwise products: $[P] \cdot r$ agrees with r on states satisfying P and is zero elsewhere, while $\langle P \rangle \cdot r$ agrees with r on states satisfying P and is ∞ elsewhere.

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The witness sequence $\{r_i\}_{i \in \mathbb{N}}$. The rule of Figure 2 uses a sequence $\{r_i\}_{i \geq 0}$ of non-negative expectations as a termination witness: intuitively, r_i is a (user-chosen) upper bound on how much of the post-expectation mass of f is still trapped inside the loop after i body executions. Premise 2 ties r_i to f at $i = 0$ by requiring $r_0 \geq f$; Premise 3 propagates the bound across one iteration of the body; and Premise 4 forces the bound — and thus the trapped mass — to vanish in the limit.

A Soundness of the While Rule

Goal. We want to show that the following rule is sound:

$$[P] \cdot f \;\leq\; \mathsf{wp}\,[\texttt{while}\ \mathsf{B}\ \texttt{do}\ \mathsf{C}]\,([P \wedge \neg\mathsf{B}] \cdot f)$$

Definitions. Define the family of approximating programs:

$$W_n \;\triangleq\; \big(\texttt{if}\ \mathsf{B}\ \texttt{then}\ \mathsf{C}\big)^n\ ;\ \texttt{if}\ \mathsf{B}\ \texttt{then}\ \texttt{abort}$$

where W_n allows at most n iterations; if the loop would continue beyond n, it aborts. Define:

$$I_n \;\triangleq\; [P] \cdot \mathsf{wp}\,[W_n]\,([P \wedge \neg\mathsf{B}] \cdot f)$$

Lemma A.1 (Recursive characterisation of I_n).

$$I_0 \;=\; [P \wedge \neg\mathsf{B}] \cdot f \qquad \textit{and} \qquad I_{n+1} \;=\; [P \wedge \mathsf{B}] \cdot \mathsf{wp}\,[\mathsf{C}]\,([P] \cdot I_n) + [P \wedge \neg\mathsf{B}] \cdot f$$

Proof. By induction on n.

Base case $n = 0$.

$$\begin{aligned} I_0 &= [P] \cdot \mathsf{wp}\,[\texttt{if}\ \mathsf{B}\ \texttt{then}\ \texttt{abort}]\,([P \wedge \neg\mathsf{B}] \cdot f) \\ &= [P] \cdot \Big([\mathsf{B}] \cdot \mathsf{wp}\,[\texttt{abort}]\,([P \wedge \neg\mathsf{B}] \cdot f) + [\neg\mathsf{B}] \cdot [P \wedge \neg\mathsf{B}] \cdot f\Big) \\ &= [P] \cdot \big([\mathsf{B}] \cdot 0 + [\neg\mathsf{B}] \cdot [P \wedge \neg\mathsf{B}] \cdot f\big) \\ &= [P] \cdot [P \wedge \neg\mathsf{B}] \cdot f \;=\; [P \wedge \neg\mathsf{B}] \cdot f \quad \checkmark \end{aligned}$$

where we used $\mathsf{wp}\,[\texttt{abort}]\,(\cdot) = 0$ and $[P] \cdot [P \wedge \neg\mathsf{B}] = [P \wedge \neg\mathsf{B}]$.

Inductive step $n \to n + 1$.

$$I_{n+1} = [P] \cdot \mathsf{wp}\,\Big[\big(\texttt{if}\ \mathsf{B}\ \texttt{then}\ \mathsf{C}\big)^{n+1}\ ;\ \texttt{if}\ \mathsf{B}\ \texttt{then}\ \texttt{abort}\Big]\,([P \wedge \neg\mathsf{B}] \cdot f)$$

Unfolding $(\texttt{if}\ \mathsf{B}\ \texttt{then}\ \mathsf{C})^{n+1} = \texttt{if}\ \mathsf{B}\ \texttt{then}\ \mathsf{C}\ ;\ (\texttt{if}\ \mathsf{B}\ \texttt{then}\ \mathsf{C})^n$ and applying the SEQ rule:

$$= [P] \cdot \mathsf{wp}\,[\texttt{if}\ \mathsf{B}\ \texttt{then}\ \mathsf{C}]\,(\mathsf{wp}\,[W_n]\,([P \wedge \neg\mathsf{B}] \cdot f))$$

By sub-induction one verifies $[P] \cdot I_k = I_k$ for all k (since $I_0 = [P \wedge \neg\mathsf{B}] \cdot f$ and each I_{k+1} carries $[P \wedge \mathsf{B}]$ or $[P \wedge \neg\mathsf{B}]$ as a factor), so in particular:

$$\mathsf{wp}\,[W_n]\,([P \wedge \neg\mathsf{B}] \cdot f) \;=\; [P] \cdot \mathsf{wp}\,[W_n]\,([P \wedge \neg\mathsf{B}] \cdot f) \;=\; I_n$$

Therefore:

$$\begin{aligned} &= [P] \cdot \mathsf{wp}\,[\texttt{if}\ \mathsf{B}\ \texttt{then}\ \mathsf{C}]\,(I_n) \\ &= [P] \cdot \big([\mathsf{B}] \cdot \mathsf{wp}\,[\mathsf{C}]\,(I_n) + [\neg\mathsf{B}] \cdot I_n\big) \\ &= [P \wedge \mathsf{B}] \cdot \mathsf{wp}\,[\mathsf{C}]\,(I_n) + [P \wedge \neg\mathsf{B}] \cdot I_n \end{aligned}$$

Using $[P] \cdot I_n = I_n$ and the identity $[P \wedge \neg\mathsf{B}] \cdot I_n = [P \wedge \neg\mathsf{B}] \cdot f$ (which follows from $[\neg\mathsf{B}] \cdot I_n = [\neg\mathsf{B}] \cdot [P \wedge \neg\mathsf{B}] \cdot f = [P \wedge \neg\mathsf{B}] \cdot f$):

$$I_{n+1} \;=\; [P \wedge \mathsf{B}] \cdot \mathsf{wp}\,[\mathsf{C}]\,([P] \cdot I_n) + [P \wedge \neg\mathsf{B}] \cdot f \quad \checkmark$$

□

Lemma A.2 (Key inequality). *Assuming premises* (a), (b), (c), *for all* $n \geq 0$*:*

$$[P] \cdot I_n + [P] \cdot r_n \;\geq\; [P] \cdot f$$

Proof. By induction on n.

Base case $n = 0$.

$$[P] \cdot I_0 + [P] \cdot r_0 \;=\; [P \wedge \neg \mathsf{B}] \cdot f + [P] \cdot r_0$$

By premise (b), $[P] \cdot r_0 \geq [P \wedge \mathsf{B}] \cdot f$, so:

$$\geq\; [P \wedge \neg \mathsf{B}] \cdot f + [P \wedge \mathsf{B}] \cdot f \;=\; [P] \cdot f \quad \checkmark$$

Inductive step $n \longrightarrow n + 1$. Expanding I_{n+1} by Theorem A.1:

$$[P] \cdot I_{n+1} + [P] \cdot r_{n+1} = [P \wedge \mathsf{B}] \cdot \mathsf{wp}\,[\mathsf{C}]\,([P] \cdot I_n) + [P \wedge \neg \mathsf{B}] \cdot f + [P] \cdot r_{n+1}$$

By premise (c), $[P] \cdot r_{n+1} \geq [P \wedge \mathsf{B}] \cdot \mathsf{wp}\,[\mathsf{C}]\,([P] \cdot r_n)$:

$$\geq [P \wedge \mathsf{B}] \cdot \mathsf{wp}\,[\mathsf{C}]\,([P] \cdot I_n) + [P \wedge \mathsf{B}] \cdot \mathsf{wp}\,[\mathsf{C}]\,([P] \cdot r_n) + [P \wedge \neg \mathsf{B}] \cdot f$$

By linearity of $\mathsf{wp}\,[\mathsf{C}]\,(\cdot)$:

$$= [P \wedge \mathsf{B}] \cdot \mathsf{wp}\,[\mathsf{C}]\,([P] \cdot I_n + [P] \cdot r_n) + [P \wedge \neg \mathsf{B}] \cdot f$$

By the induction hypothesis $[P] \cdot I_n + [P] \cdot r_n \geq [P] \cdot f$ and monotonicity of $\mathsf{wp}\,[\mathsf{C}]\,(\cdot)$:

$$\geq [P \wedge \mathsf{B}] \cdot \mathsf{wp}\,[\mathsf{C}]\,([P] \cdot f) + [P \wedge \neg \mathsf{B}] \cdot f$$

By premise (a), $[P \wedge \mathsf{B}] \cdot \mathsf{wp}\,[\mathsf{C}]\,([P] \cdot f) \geq [P \wedge \mathsf{B}] \cdot f$:

$$\geq\; [P \wedge \mathsf{B}] \cdot f + [P \wedge \neg \mathsf{B}] \cdot f \;=\; [P] \cdot f \quad \checkmark$$

□

Main result. From Theorem A.2, for all $n \geq 0$:

$$[P] \cdot I_n + [P] \cdot r_n \;\geq\; [P] \cdot f$$

Taking the limit:

$$\lim_n([P] \cdot I_n) + \lim_n([P] \cdot r_n) \;\geq\; [P] \cdot f$$

By condition (d), $\lim_n r_n = 0$, so $\lim_n([P] \cdot r_n) = 0$:

$$\lim_n([P] \cdot I_n) \;\geq\; [P] \cdot f$$

By Scott-continuity and Theorem A.1, $\lim_n I_n = \mathsf{wp}\,[\texttt{while}\ \mathsf{B}\ \texttt{do}\ \mathsf{C}]\,([P \wedge \neg \mathsf{B}] \cdot f)$, giving:

$$\mathsf{wp}\,[\texttt{while}\ \mathsf{B}\ \texttt{do}\ \mathsf{C}]\,([P \wedge \neg \mathsf{B}] \cdot f) \;\geq\; [P] \cdot f$$

which is exactly the desired conclusion. □

B Relative Completeness

Theorem: Relative Completeness. **Assumptions:**

1. $[P] \cdot f \;\leq\; \mathsf{wp}\,[\texttt{while}\ \mathsf{B}\ \texttt{do}\ \mathsf{C}]\,([\neg \mathsf{B} \wedge P] \cdot f)$
2. $[\mathsf{B} \wedge P] \cdot f \;\leq\; \mathsf{wp}\,[\mathsf{C}]\,([P] \cdot f)$

Definitions:

$$\begin{aligned} H &\triangleq \mathsf{wp}\,[\texttt{while}\ \mathsf{B}\ \texttt{do}\ \mathsf{C}]\,([\neg \mathsf{B} \wedge P] \cdot f) \\ I_0 &\triangleq [\neg \mathsf{B} \wedge P] \cdot f \\ I_{i+1} &\triangleq [\mathsf{B}] \cdot \mathsf{wp}\,[\mathsf{C}]\,(I_i) + I_0 \\ r_i &\triangleq H - I_i \end{aligned}$$

Verification of Premise 2:

$$\begin{aligned} \mathsf{B} \wedge P &\implies I_0 = 0 \\ r_0 &= H - I_0 = H \\ \text{From (1): } H \geq [P] \cdot f &\implies H \geq f \\ [\mathsf{B} \wedge P] \cdot r_0 &\geq [\mathsf{B} \wedge P] \cdot f \end{aligned}$$

Verification of Premise 2:

$$\begin{aligned} &\mathsf{B} \wedge P \implies I_0 = 0 \\ &r_0 = H - I_0 = H \\ &\text{P holds so: } H \geq [P] \cdot f \implies H \geq f \\ &[\mathsf{B} \wedge P] \cdot r_0 \geq [\mathsf{B} \wedge P] \cdot f \end{aligned}$$

Verification of Premise 3:

$$\begin{aligned} H &= [\mathsf{B}] \cdot \mathrm{wp}\,[\mathsf{C}]\,(H) + I_0 \\ r_{i+1} &= H - I_{i+1} \\ r_{i+1} &= ([\mathsf{B}] \cdot \mathrm{wp}\,[\mathsf{C}]\,(H) + I_0) - ([\mathsf{B}] \cdot \mathrm{wp}\,[\mathsf{C}]\,(I_i) + I_0) \\ r_{i+1} &= [\mathsf{B}] \cdot (\mathrm{wp}\,[\mathsf{C}]\,(H) - \mathrm{wp}\,[\mathsf{C}]\,(I_i)) \\ r_{i+1} &= [\mathsf{B}] \cdot \mathrm{wp}\,[\mathsf{C}]\,(H - I_i) = [\mathsf{B}] \cdot \mathrm{wp}\,[\mathsf{C}]\,(r_i) \\ \text{By Monotonicity: } r_i \geq [P] \cdot r_i &\implies \mathrm{wp}\,[\mathsf{C}]\,(r_i) \geq \mathrm{wp}\,[\mathsf{C}]\,([P] \cdot r_i) \\ [\mathsf{B} \wedge P] \cdot r_{i+1} &\geq [\mathsf{B} \wedge P] \cdot \mathrm{wp}\,[\mathsf{C}]\,([P] \cdot r_i) \end{aligned}$$

Verification of Premise 4:

$$\begin{aligned} &\lim_{i \to \infty} I_i = H \\ &\lim_{i \to \infty} r_i = H - \lim_{i \to \infty} I_i = H - H = 0 \\ &\lim_{i \to \infty} r_i = 0 \end{aligned}$$

C Derived Rules

C.1 Multiplicative decay

$$\frac{\{\mathsf{B} \wedge P \mid f\}_{\leq}\ \mathsf{C}\ \{P \mid f\} \quad \mathsf{B} \wedge P \vDash V \geq f \quad \{\mathsf{B} \wedge P \mid \beta \cdot V\}_{\geq}\ \mathsf{C}\ \{\top \mid [P] \cdot V\} \quad \beta \in [0,1) \quad \mathsf{B} \wedge P \vDash V < \infty}{\{P \mid f\}_{\leq}\ \texttt{while}\ \mathsf{B}\ \texttt{do}\ \mathsf{C}\ \{\neg\mathsf{B} \wedge P \mid f\}}\ [\text{WHILE-MULT}]$$

C.1.1 Proof via Main Rule Soundness.

Premises:

$$\begin{aligned} &(1)\ [\mathsf{B} \wedge P] \cdot f < \mathrm{wp}\,[\mathsf{C}]\,([P] \cdot f) \\ &(2)\ V \geq f \\ &(3)\ \mathrm{wp}\,[\mathsf{C}]\,([P] \cdot V) < \beta \cdot V, \quad \beta < 1 \\ &(4)\ V < \infty \end{aligned}$$

Proof Strategy: $r_i \triangleq \beta^i \cdot V$.

Verification of Main Rule Premise 2:

$$r_0 = V \xrightarrow{\text{by (2)}} r_0 \geq f \implies [\mathsf{B} \wedge P] \cdot r_0 \geq [\mathsf{B} \wedge P] \cdot f$$

Verification of Main Rule Premise 3:

$$\begin{aligned} \text{wp}\,[\mathsf{C}]\,([P] \cdot r_i) &= \text{wp}\,[\mathsf{C}]\,\left([P] \cdot \beta^i \cdot V\right) = \beta^i \cdot \text{wp}\,[\mathsf{C}]\,([P] \cdot V) \\ &< \beta^i \cdot (\beta \cdot V) \quad \text{(by (3))} \\ &= \beta^{i+1} \cdot V = r_{i+1} \\ \implies [\mathsf{B} \wedge P] \cdot r_{i+1} &\geq [\mathsf{B} \wedge P] \cdot \text{wp}\,[\mathsf{C}]\,([P] \cdot r_i) \end{aligned}$$

Verification of Main Rule Premise 4:

$$\lim_{i \to \infty} r_i = \lim_{i \to \infty} (\beta^i \cdot V) \xrightarrow{\beta < 1, V < \infty} 0 \implies \lim_{i \to \infty} [\mathsf{B} \wedge P] \cdot r_i = 0$$

Conclusion: Soundness follows from the main rule. ✓

C.1.2 Direct Proof.

Definitions: (Recall I_n from Theorem A.1)

$$r_i \triangleq \beta^i \cdot V$$

Auxiliary inequality on $\mathsf{B} \wedge P$:

$$\begin{aligned} [\mathsf{B} \wedge P] \cdot \beta \cdot V &\geq \text{wp}\,[\mathsf{C}]\,([P] \cdot V) \quad \text{(premise 3)} \\ \beta^n \cdot [\mathsf{B} \wedge P] \cdot \beta \cdot V &\geq \beta^n \cdot \text{wp}\,[\mathsf{C}]\,([P] \cdot V) \\ [\mathsf{B} \wedge P] \cdot r_{n+1} &\geq \text{wp}\,[\mathsf{C}]\,([P] \cdot r_n) \quad \text{(by linearity)} \end{aligned}$$

Key inequality: $\forall n.\ [P] \cdot I_n + [P] \cdot r_n \geq [P] \cdot f$ (by induction)

Base case $n = 0$:

$$\begin{aligned} [P] \cdot I_0 + [P] \cdot r_0 &= [\neg\mathsf{B} \wedge P] \cdot f + [P] \cdot V \\ &\geq [\neg\mathsf{B} \wedge P] \cdot f + [\mathsf{B} \wedge P] \cdot V \\ &\geq [\neg\mathsf{B} \wedge P] \cdot f + [\mathsf{B} \wedge P] \cdot f \quad \text{(by premise 2)} \\ &= [P] \cdot f \end{aligned}$$

Inductive step $n \to n+1$ (using Theorem A.1 for I_{n+1}):

$$\begin{aligned} [P] \cdot I_{n+1} + [P] \cdot r_{n+1} &= [\mathsf{B} \wedge P] \cdot \text{wp}\,[\mathsf{C}]\,([P] \cdot I_n) + [\neg\mathsf{B} \wedge P] \cdot f + [P] \cdot r_{n+1} \\ &\geq [\mathsf{B} \wedge P] \cdot \text{wp}\,[\mathsf{C}]\,([P] \cdot I_n) + [\neg\mathsf{B} \wedge P] \cdot f + [\mathsf{B} \wedge P] \cdot \text{wp}\,[\mathsf{C}]\,([P] \cdot r_n) \\ &= [\mathsf{B} \wedge P] \cdot \text{wp}\,[\mathsf{C}]\,([P] \cdot I_n + [P] \cdot r_n) + [\neg\mathsf{B} \wedge P] \cdot f \quad \text{(by linearity)} \\ &\geq [\mathsf{B} \wedge P] \cdot \text{wp}\,[\mathsf{C}]\,([P] \cdot f) + [\neg\mathsf{B} \wedge P] \cdot f \quad \text{(by IH + monotonicity)} \\ &\geq [\mathsf{B} \wedge P] \cdot f + [\neg\mathsf{B} \wedge P] \cdot f \quad \text{(by premise 1)} \\ &= [P] \cdot f \end{aligned}$$

Limit argument:

$$\begin{aligned} \beta \in [0, 1),\ \mathsf{B} \wedge P \implies V < \infty \implies \lim_{n \to \infty} r_n &= 0 \\ \lim_{n \to \infty} I_n &= \text{wp}\,[\texttt{while}\ \mathsf{B}\ \texttt{do}\ \mathsf{C}]\,([\neg\mathsf{B} \wedge P] \cdot f) \quad \text{(by Scott-continuity)} \\ \implies [P] \cdot f &\leq \text{wp}\,[\texttt{while}\ \mathsf{B}\ \texttt{do}\ \mathsf{C}]\,([\neg\mathsf{B} \wedge P] \cdot f) \quad ✓ \end{aligned}$$

C.2 Additive decrease

Rule.

$$\frac{\{\mathsf{B} \wedge P \mid f\}_{\leq}\ \mathsf{C}\ \{P \mid f\} \quad \mathsf{B} \wedge P \vDash V \geq f \quad [\mathsf{B} \wedge P] \cdot V \geq \mathsf{wp}\,[\mathsf{C}]\,([P] \cdot V) + \epsilon \quad \epsilon > 0 \quad \mathsf{B} \wedge P \vDash V \leq M}{\{P \mid f\}_{\leq}\ \mathtt{while}\ \mathsf{B}\ \mathtt{do}\ \mathsf{C}\ \{\neg\mathsf{B} \wedge P \mid f\}}\ [\text{WHILE-ADD}]$$

C.2.1 *Proof via Main Rule Soundness.*

Premises:

$$(1)\ [\mathsf{B} \wedge P] \cdot f < \mathsf{wp}\,[\mathsf{C}]\,([P] \cdot f)$$
$$(2)\ V \geq f$$
$$(3)\ \mathsf{wp}\,[\mathsf{C}]\,([P] \cdot V) < V - \epsilon$$
$$(4)\ \epsilon > 0,\ V \leq M$$

Proof Strategy: $r_i \triangleq \max(0, V - i \cdot \epsilon)$.

Verification of Main Rule Premise 2:

$$r_0 = V \xrightarrow{\text{by (2)}} r_0 \geq f \implies [\mathsf{B} \wedge P] \cdot r_0 \geq [\mathsf{B} \wedge P] \cdot f$$

Verification of Main Rule Premise 3:

$$\begin{aligned}
&\text{Assume } V - i\epsilon > 0 \text{ (otherwise } r_i = 0 \text{ trivially):}\\
&\mathsf{wp}\,[\mathsf{C}]\,([P] \cdot r_i) < \mathsf{wp}\,[\mathsf{C}]\,([P] \cdot (V - i\epsilon)) = \mathsf{wp}\,[\mathsf{C}]\,([P] \cdot V) - i\epsilon\\
&\qquad < (V - \epsilon) - i\epsilon \quad \text{(by (3))}\\
&\qquad = V - (i+1)\epsilon < r_{i+1}\\
&\implies [\mathsf{B} \wedge P] \cdot r_{i+1} \geq [\mathsf{B} \wedge P] \cdot \mathsf{wp}\,[\mathsf{C}]\,([P] \cdot r_i)
\end{aligned}$$

Verification of Main Rule Premise 4:

$$\begin{aligned}
&\text{Since } \epsilon > 0 \text{ and } V \leq M < \infty,\ \exists N \text{ s.t. } \forall i > N : V - i\epsilon < 0\\
&\therefore \lim_{i \to \infty} r_i = 0 \implies \lim_{i \to \infty} [\mathsf{B} \wedge P] \cdot r_i = 0
\end{aligned}$$

Conclusion: Soundness follows from the main rule. ✓

C.2.2 *Direct Proof.*

Definitions: (Recall I_n from Theorem A.1)

$$\beta \triangleq 1 - \epsilon/M$$
$$r_i \triangleq \beta^i \cdot V$$

Range of β:

$$\begin{aligned}
0 \leq \mathsf{wp}\,[\mathsf{C}]\,([P] \cdot V) \leq V - \epsilon \leq M - \epsilon &\implies \epsilon \leq M\\
0 < \epsilon \leq M &\implies \beta \in [0, 1)
\end{aligned}$$

Auxiliary inequality on $\mathsf{B} \wedge P$:

$$\begin{aligned}
&V \leq M \implies V \cdot \epsilon/M \leq \epsilon \implies V - \epsilon \leq \beta \cdot V\\
&\mathsf{wp}\,[\mathsf{C}]\,([P] \cdot V) \leq V - \epsilon \implies \mathsf{wp}\,[\mathsf{C}]\,([P] \cdot V) \leq \beta \cdot V \quad \text{(by premise 3)}\\
&\beta^n \cdot \mathsf{wp}\,[\mathsf{C}]\,([P] \cdot V) \leq \beta^{n+1} \cdot V\\
&\mathsf{wp}\,[\mathsf{C}]\,([P] \cdot r_n) \leq r_{n+1} \quad \text{(by linearity)}\\
&\implies [\mathsf{B} \wedge P] \cdot r_{n+1} \geq \mathsf{wp}\,[\mathsf{C}]\,([P] \cdot r_n)
\end{aligned}$$

Key inequality: $\forall n.\ [P] \cdot I_n + [P] \cdot r_n \geq [P] \cdot f$ (by induction)

Base case $n = 0$:

$$\begin{aligned}
[P] \cdot I_0 + [P] \cdot r_0 &= [\neg \mathsf{B} \wedge P] \cdot f + [P] \cdot V \\
&\geq [\neg \mathsf{B} \wedge P] \cdot f + [\mathsf{B} \wedge P] \cdot V \\
&\geq [\neg \mathsf{B} \wedge P] \cdot f + [\mathsf{B} \wedge P] \cdot f \quad \text{(by premise 2)} \\
&= [P] \cdot f
\end{aligned}$$

Inductive step $n \longrightarrow n + 1$ (using Theorem A.1 for I_{n+1}):

$$\begin{aligned}
[P] \cdot I_{n+1} + [P] \cdot r_{n+1} &= [\mathsf{B} \wedge P] \cdot \mathsf{wp}\,[\mathsf{C}]\,([P] \cdot I_n) + [\neg \mathsf{B} \wedge P] \cdot f + [P] \cdot r_{n+1} \\
&\geq [\mathsf{B} \wedge P] \cdot \mathsf{wp}\,[\mathsf{C}]\,([P] \cdot I_n) + [\neg \mathsf{B} \wedge P] \cdot f + [\mathsf{B} \wedge P] \cdot \mathsf{wp}\,[\mathsf{C}]\,([P] \cdot r_n) \\
&= [\mathsf{B} \wedge P] \cdot \mathsf{wp}\,[\mathsf{C}]\,([P] \cdot I_n + [P] \cdot r_n) + [\neg \mathsf{B} \wedge P] \cdot f \quad \text{(by linearity)} \\
&\geq [\mathsf{B} \wedge P] \cdot \mathsf{wp}\,[\mathsf{C}]\,([P] \cdot f) + [\neg \mathsf{B} \wedge P] \cdot f \quad \text{(by IH + monotonicity)} \\
&\geq [\mathsf{B} \wedge P] \cdot f + [\neg \mathsf{B} \wedge P] \cdot f \quad \text{(by premise 1)} \\
&= [P] \cdot f
\end{aligned}$$

Limit argument:

$$\begin{aligned}
\beta \in [0, 1),\ V \leq M \implies & \lim_{n \to \infty} r_n = 0 \\
& \lim_{n \to \infty} I_n = \mathsf{wp}\,[\texttt{while}\ \mathsf{B}\ \texttt{do}\ \mathsf{C}]\,([\neg \mathsf{B} \wedge P] \cdot f) \quad \text{(by Scott-continuity)} \\
\implies & [P] \cdot f \leq \mathsf{wp}\,[\texttt{while}\ \mathsf{B}\ \texttt{do}\ \mathsf{C}]\,([\neg \mathsf{B} \wedge P] \cdot f) \quad \checkmark
\end{aligned}$$